

Problem set 1

Solution Key

Econ 5030

1 Preferences and Utility

Throughout, $x \succ y$ means ($x \succsim y$ and not $y \succsim x$), and $x \sim y$ means ($x \succsim y$ and $y \succsim x$).

1. Assume that $\succsim$ is complete and transitive on X .
 - (a) $x \succ x$ *never holds*. By definition, $x \succ x$ would require $x \succsim x$ and not $x \succsim x$ simultaneously, which is a contradiction. Hence $x \succ x$ never holds. (Completeness in fact gives $x \succsim x$ for every x , so the second requirement always fails.)
 - (b) *If $x \succ y$ and $y \succ z$, then $x \succ z$.* From $x \succ y$ and $y \succ z$ we have $x \succsim y$ and $y \succsim z$, so $x \succsim z$ by transitivity. It remains to show that $z \succsim x$ fails. Suppose $z \succsim x$. Together with $x \succsim y$, transitivity gives $z \succsim y$, contradicting $y \succ z$. Therefore not $z \succsim x$, and $x \succ z$.
 - (c) $x \sim x$ *for all x* . By completeness (applied to the pair x, x), $x \succsim x$. Then $x \succsim x$ and $x \succsim x$, which is exactly $x \sim x$.
 - (d) *If $x \sim y$ and $y \sim z$, then $x \sim z$.* We have $x \succsim y$, $y \succsim x$, $y \succsim z$, $z \succsim y$. Transitivity on $x \succsim y$, $y \succsim z$ gives $x \succsim z$; transitivity on $z \succsim y$, $y \succsim x$ gives $z \succsim x$. Hence $x \sim z$.
 - (e) *If $x \sim y$, then $y \sim x$.* $x \sim y$ means $x \succsim y$ and $y \succsim x$. This is the same pair of statements as $y \succsim x$ and $x \succsim y$, i.e. $y \sim x$.
 - (f) *If $x \succ y$ and $y \succ z$, then $x \succ z$.* From $x \succ y$ and $y \succ z$, transitivity gives $x \succ z$. Suppose $z \succsim x$. Then $y \succ z$ and $z \succsim x$ give $y \succ x$ by transitivity, contradicting $x \succ y$. Hence not $z \succsim x$, so $x \succ z$.
2. $X = \{a, b, c, d\}$ and $\succsim$ is complete and transitive on X .
 - (a) $C(X, \succsim)$ *is non-empty*. Build a best element by pairwise comparison. By completeness, either $a \succsim b$ or $b \succsim a$; let $x_2 \in \{a, b\}$ denote the (weakly) preferred one, so $x_2 \succsim a$ and $x_2 \succsim b$ (using $x_2 \succsim x_2$ from completeness). By completeness again, compare x_2 with c and let $x_3 \in \{x_2, c\}$ be the preferred one. If $x_3 = x_2$ then $x_3 \succsim a, b, c$ directly. If $x_3 = c$, then $c \succsim x_2$, and $x_2 \succsim a$, $x_2 \succsim b$ give $c \succsim a$ and $c \succsim b$ by transitivity; also $c \succsim c$. Either way $x_3 \succsim y$ for all $y \in \{a, b, c\}$. Repeat

once more with d to obtain x_4 with $x_4 \succsim y$ for all $y \in X$. Then $x_4 \in C(X, \succsim)$, so $C(X, \succsim) \neq \emptyset$.

(The same induction works for any finite X . Alternatively: if $C(X, \succsim) = \emptyset$, then for every x some y satisfies $y \succ x$ (by completeness, “not $x \succsim y$ ” implies $y \succ x$). Starting anywhere, this produces a chain $x_1 \prec x_2 \prec x_3 \prec \dots$; with only four elements some element repeats, and transitivity of $\succ$ (part 1(b)) then yields $x_i \succ x_i$, contradicting 1(a).)

- (b) $x \sim y$ for all $x, y \in C(X, \succsim)$. Let $x, y \in C(X, \succsim)$. Since $x \in C(X, \succsim)$, $x \succsim z$ for every $z \in X$; in particular $x \succsim y$. Since $y \in C(X, \succsim)$, $y \succsim x$. Hence $x \sim y$.

3. $X = \{a, b, c, d\}$.

- (a) *Complete and transitive $\succsim$ on X has a utility representation.* Define, for each $x \in X$,

$$u(x) = |\{z \in X : x \succsim z\}|,$$

the number of alternatives that x is weakly preferred to. Let $L(x) = \{z \in X : x \succsim z\}$, so $u(x) = |L(x)|$. We show $x \succsim y \iff u(x) \geq u(y)$.

($\Rightarrow$) Suppose $x \succsim y$. If $z \in L(y)$ then $y \succsim z$, and transitivity gives $x \succsim z$, i.e. $z \in L(x)$. Thus $L(y) \subseteq L(x)$ and $u(y) \leq u(x)$.

($\Leftarrow$) Suppose $u(x) \geq u(y)$ but not $x \succsim y$. By completeness $y \succsim x$, so $y \succ x$. By the argument just given, $L(x) \subseteq L(y)$. Moreover $y \in L(y)$ (completeness gives $y \succsim y$) while $y \notin L(x)$ (since not $x \succsim y$). So $L(x) \subsetneq L(y)$ and $u(x) < u(y)$, a contradiction. Hence $x \succsim y$.

Therefore $u : X \rightarrow \mathbb{R}$ represents $\succsim$. (Only finiteness of X is used, so this proves the result for any finite set.)

- (b) *Why an extra assumption is needed on $X' = [0, 1]^2$.* The construction above relies on X being finite (or, more generally, countable) so that counting, or an enumeration, produces real numbers in the right order. When X' is uncountable, completeness and transitivity alone do not guarantee a representation: the preference may have “too many” strictly ordered levels to fit into $\mathbb{R}$ in an order-preserving way. The standard additional assumption is *continuity* (for every x , the sets $\{y : y \succsim x\}$ and $\{y : x \succsim y\}$ are closed), which, together with completeness and transitivity, guarantees a (continuous) utility representation (Debreu’s theorem).

Example (lexicographic preferences). On $X' = [0, 1]^2$ define

$$(x_1, x_2) \succsim (y_1, y_2) \iff [x_1 > y_1] \text{ or } [x_1 = y_1 \text{ and } x_2 \geq y_2].$$

This is complete (for any two bundles either the first coordinates differ, or they agree and the second coordinates are comparable) and transitive (check the cases: if $x_1 > y_1 \geq z_1$ or $x_1 \geq y_1 > z_1$ then $x_1 > z_1$; if $x_1 = y_1 = z_1$ then $x_2 \geq y_2 \geq z_2$).

Suppose u represents it. For each $t \in [0, 1]$, $(t, 1) \succ (t, 0)$, so $u(t, 1) > u(t, 0)$ and the open interval $I_t = (u(t, 0), u(t, 1))$ is non-empty. If $t < t'$, then $(t', 0) \succ (t, 1)$, so $u(t', 0) > u(t, 1)$ and $I_t \cap I_{t'} = \emptyset$. Thus $\{I_t\}_{t \in [0, 1]}$ is an uncountable family of pairwise disjoint non-empty open intervals in $\mathbb{R}$. Each contains a rational number q_t ,

and $t \mapsto q_t$ is injective, giving an injection from $[0, 1]$ into $\mathbb{Q}$. This contradicts the countability of $\mathbb{Q}$. Hence no utility function represents lexicographic preferences. (Lexicographic preferences fail continuity: the upper contour set of $(1/2, 1/2)$ is not closed.)

4. Cobb–Douglas: $u(x_1, x_2) = x_1^{1/2} x_2^{1/2}$ on $\mathbb{R}_+^2$, $p \in \mathbb{R}_{++}^2$, $w > 0$.

(a) *Interior and budget-exhausting.* If $x_1 = 0$ or $x_2 = 0$ then $u(x) = 0$. The bundle $(\frac{w}{2p_1}, \frac{w}{2p_2})$ is affordable (it costs exactly w) and yields $u = \frac{w}{2\sqrt{p_1 p_2}} > 0$. Hence any optimal bundle has $x_1 > 0$ and $x_2 > 0$, i.e. is interior.

u is strictly increasing in each argument on $\mathbb{R}_{++}^2$. If $p \cdot x < w$ at an interior x , then $x' = (x_1 + \varepsilon, x_2)$ is affordable for small $\varepsilon > 0$ and $u(x') > u(x)$, so x is not optimal. Hence the optimum satisfies $p_1 x_1 + p_2 x_2 = w$.

First-order conditions. Since the optimum is interior with a binding budget, the Lagrangian $\mathcal{L} = x_1^{1/2} x_2^{1/2} - \lambda(p_1 x_1 + p_2 x_2 - w)$ gives

$$\frac{1}{2} x_1^{-1/2} x_2^{1/2} = \lambda p_1, \quad \frac{1}{2} x_1^{1/2} x_2^{-1/2} = \lambda p_2, \quad p_1 x_1 + p_2 x_2 = w.$$

Dividing the first two conditions,

$$\frac{x_2}{x_1} = \frac{p_1}{p_2} \implies p_2 x_2 = p_1 x_1.$$

Substituting into the budget: $2p_1 x_1 = w$. Hence the Marshallian demand is

$$D(p, w) = \left(\frac{w}{2p_1}, \frac{w}{2p_2} \right).$$

(Each good receives exactly half of income, the Cobb–Douglas exponent share.) The FOCs are sufficient because $\ln u = \frac{1}{2} \ln x_1 + \frac{1}{2} \ln x_2$ is strictly concave on $\mathbb{R}_{++}^2$ and the budget set is convex, so the maximizer is unique. The indirect utility function is

$$v(p, w) = \left(\frac{w}{2p_1} \right)^{1/2} \left(\frac{w}{2p_2} \right)^{1/2} = \frac{w}{2\sqrt{p_1 p_2}}.$$

(b) $(p_1, p_2, w) = (4, 1, 80)$.

$$x_1 = \frac{80}{2 \cdot 4} = 10, \quad x_2 = \frac{80}{2 \cdot 1} = 40, \quad v(4, 1, 80) = \frac{80}{2\sqrt{4 \cdot 1}} = 20.$$

Check: $u(10, 40) = \sqrt{10 \cdot 40} = \sqrt{400} = 20$.

(c) *Walras' law and homogeneity of degree zero with $\lambda = 2$.* Walras' law: $p \cdot D(p, w) = 4 \cdot 10 + 1 \cdot 40 = 80 = w$. ✓

Homogeneity: $(\lambda p, \lambda w) = (8, 2, 160)$, and

$$D(8, 2, 160) = \left(\frac{160}{2 \cdot 8}, \frac{160}{2 \cdot 2} \right) = (10, 40) = D(4, 1, 80). \quad \checkmark$$

In general $D(\lambda p, \lambda w) = \left(\frac{\lambda w}{2\lambda p_1}, \frac{\lambda w}{2\lambda p_2} \right) = D(p, w)$, and likewise $v(8, 2, 160) = \frac{160}{2\sqrt{16}} = 20 = v(4, 1, 80)$.

5. Perfect complements: $u(x_1, x_2) = \min\{x_1, 2x_2\}$ on $\mathbb{R}_+^2$, $p \in \mathbb{R}_{++}^2$, $w \geq 0$.

- (a) *Marshallian demand and indirect utility.* Take any affordable bundle x and let $t = \min\{x_1, 2x_2\}$. Then $x_1 \geq t$ and $x_2 \geq t/2$, so

$$w \geq p_1 x_1 + p_2 x_2 \geq p_1 t + p_2 \frac{t}{2} = \frac{t(2p_1 + p_2)}{2} \implies t \leq \frac{2w}{2p_1 + p_2}.$$

So no affordable bundle gives utility above $\frac{2w}{2p_1 + p_2}$. The bundle

$$x^* = \left(\frac{2w}{2p_1 + p_2}, \frac{w}{2p_1 + p_2} \right)$$

costs $p_1 \frac{2w}{2p_1 + p_2} + p_2 \frac{w}{2p_1 + p_2} = w$ and satisfies $x_1^* = 2x_2^*$, so $u(x^*) = \frac{2w}{2p_1 + p_2}$, attaining the bound. It is the unique maximizer: attaining the bound requires every inequality above to hold with equality, i.e. $x_1 = t$, $x_2 = t/2$ (no “wasted” units of either good) and $p \cdot x = w$, which pins down $x = x^*$. Therefore

$$D(p, w) = \left(\frac{2w}{2p_1 + p_2}, \frac{w}{2p_1 + p_2} \right), \quad v(p, w) = \frac{2w}{2p_1 + p_2}.$$

Economic explanation. The goods are consumed in the fixed proportion $x_1 = 2x_2$: two units of good 1 are needed for each unit of good 2, and any unit beyond that ratio adds nothing to utility. The consumer therefore buys “kits” of $(2, 1)$, each costing $2p_1 + p_2$; with income w the number of kits is $w/(2p_1 + p_2)$, which is exactly $v(p, w)/2$. Relative prices do not change the proportions (there is no substitution between the goods); prices only affect how many kits are affordable.

- (b) $(p_1, p_2, w) = (3, 2, 80)$. $2p_1 + p_2 = 8$, so

$$x_1 = \frac{2 \cdot 80}{8} = 20, \quad x_2 = \frac{80}{8} = 10, \quad v(3, 2, 80) = \frac{160}{8} = 20.$$

Check: $3 \cdot 20 + 2 \cdot 10 = 80$ and $\min\{20, 2 \cdot 10\} = 20$.

- (c) *Convex but not strictly convex; demand is still a singleton.*

Convex. For any t , the upper contour set is

$$\{x \in \mathbb{R}_+^2 : \min\{x_1, 2x_2\} \geq t\} = \{x_1 \geq t\} \cap \{x_2 \geq t/2\},$$

an intersection of two half-planes, hence convex. So $\succsim$ is convex. (Equivalently, u is the minimum of two linear functions, hence concave, hence quasi-concave.)

Not strictly convex. Take $x = (2, 1)$ and $y = (4, 1)$. Then $u(x) = \min\{2, 2\} = 2$ and $u(y) = \min\{4, 2\} = 2$, so $x \sim y$ with $x \neq y$. The midpoint $z = \frac{1}{2}x + \frac{1}{2}y = (3, 1)$ has $u(z) = \min\{3, 2\} = 2$, so $z \sim x$ rather than $z \succ x$. Strict convexity would require $z \succ x$, so it fails: indifference curves are L-shaped and contain flat segments.

Why demand is nonetheless a singleton. Strict convexity is sufficient, not necessary, for single-valued demand. The flat parts of an indifference curve are bundles with $x_1 \neq 2x_2$, and on any such bundle some expenditure is wasted; by the argument in

(a), such a bundle can be affordably replaced by one on a strictly higher indifference curve (when $w > 0$), so it is never optimal. Only the kink $x_1 = 2x_2$ can be optimal, and the budget line meets that ray at exactly one point. Hence $D(p, w)$ is the single bundle found in (a) (and $D(p, 0) = \{(0, 0)\}$ when $w = 0$). A singleton is trivially a convex set, consistent with the general result that convex preferences yield convex-valued demand.